Tutorial 6: Predicting Top 5 Labels with Pretrained Models (ResNet-50 and VGG-16)

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1. Objective

The goal of this experiment was to evaluate and compare the predictive accuracy and confidence of several pre-trained convolutional neural network (CNN) architectures using the ImageNet dataset categories.

2. Methodology

- **Input Image:** A local image file (dog.jpeg) with the ground truth label **Golden Retriever**.
- **Preprocessing:** All images were resized to 256 pixels, center-cropped to 224x224, converted to tensors, and normalized using standard ImageNet mean and standard deviation values.
- **Environment:** The models were executed on a CPU-based environment using PyTorch and torchvision.

3. Model Performance Summary

Architecture	Top Prediction	Confidence (%)	Result
VGG-16	Golden Retriever	91.92%	**Correct**
ResNet-50	Golden Retriever	44.66%	**Correct**
AlexNet	Cocker Spaniel	75.00%	**Incorrect**
ResNet-101	Golden Retriever	76.93%	**Correct**
MobileNet_V2	Golden Retriever	10.70%	**Correct**

4. Detailed Analysis & Screenshot Placeholders

A. VGG-16 Performance

VGG-16 showed the highest confidence level for the correct class, suggesting a very strong feature match for this specific image.

```python
vgg16 = models.vgg16(weights='DEFAULT').to(device).eval()
evaluate_model(vgg16, "VGG-16", ground_truth)
```

[86] ✓ 0.9s Python

```
ARCHITECTURE: VGG-16
----------------------------
TOP 5 PREDICTIONS:
1. golden retriever: 91.92%
2. Sussex spaniel: 1.98%
3. Irish setter: 0.89%
4. Labrador retriever: 0.76%
5. cocker spaniel: 0.61%
----------------------------
ACTUAL LABEL:    Golden Retriever
PREDICTED LABEL: golden retriever
RESULT: CORRECT
```

B. ResNet Series (50 vs 101)

Both ResNet models correctly identified the image. Interestingly, the deeper ResNet-101 showed significantly higher confidence (76.93%) compared to ResNet-50 (44.66%).

```python
resnet50 = models.resnet50(weights='DEFAULT').to(device).eval()
evaluate_model(resnet50, "ResNet-50", ground_truth)
```

[87] ✓ 0.4s Python

```
ARCHITECTURE: RESNET-50
----------------------------
TOP 5 PREDICTIONS:
1. golden retriever: 44.66%
2. tennis ball: 1.07%
3. flat-coated retriever: 0.99%
4. Irish setter: 0.52%
5. Brittany spaniel: 0.42%
----------------------------
ACTUAL LABEL:    Golden Retriever
PREDICTED LABEL: golden retriever
RESULT: CORRECT
```

```python
resnet101 = models.resnet101(weights='DEFAULT').to(device).eval()
evaluate_model(resnet101, "ResNet-101", ground_truth)
```

[89] ✓ 1.3s Python

```
ARCHITECTURE: RESNET-101
----------------------------
TOP 5 PREDICTIONS:
1. golden retriever: 76.93%
2. tennis ball: 1.38%
3. Irish setter: 0.77%
4. Brittany spaniel: 0.34%
5. Labrador retriever: 0.31%
----------------------------
ACTUAL LABEL:    Golden Retriever
PREDICTED LABEL: golden retriever
RESULT: CORRECT
```

C. AlexNet (The Outlier)

AlexNet was the only model to misclassify the image as a "cocker spaniel." This is likely due to its simpler architecture and older design compared to the more modern residual or depth-wise separable networks.

```python
alexnet = models.alexnet(weights='DEFAULT').to(device).eval()
evaluate_model(alexnet, "AlexNet", ground_truth)
```

[88] ✓ 0.3s Python

```
ARCHITECTURE: ALEXNET
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TOP 5 PREDICTIONS:
1. cocker spaniel: 75.00%
2. golden retriever: 23.76%
3. Sussex spaniel: 0.50%
4. tennis ball: 0.38%
5. Irish setter: 0.25%
------------------------------
ACTUAL LABEL:    Golden Retriever
PREDICTED LABEL: cocker spaniel
RESULT: INCORRECT
```

D. MobileNet_V2

While MobileNet_V2 was correct, its confidence was extremely low (10.70%). This reflects the trade-off between model size/efficiency and classification certainty.

```python
mobilenet = models.mobilenet_v2(weights='DEFAULT').to(device).eval()
evaluate_model(mobilenet, "MobileNet_V2", ground_truth)
```

[90] ✓ 0.2s Python

```
ARCHITECTURE: MOBILENET_V2
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TOP 5 PREDICTIONS:
1. golden retriever: 10.70%
2. cocker spaniel: 5.21%
3. Irish setter: 2.97%
4. Sussex spaniel: 1.37%
5. clumber: 0.74%
------------------------------
ACTUAL LABEL:    Golden Retriever
PREDICTED LABEL: golden retriever
RESULT: CORRECT
```

5. Key Observations

- **Accuracy vs. Complexity:** Newer and deeper models generally performed better, though VGG-16 (an older but very "heavy" model) excelled at this specific high-resolution feature extraction.
- **Confidence Levels:** High confidence does not always guarantee accuracy (as seen in AlexNet's 75% confidence in a wrong answer), but deeper ResNet architectures showed a clear improvement in "certainty" as depth increased.
- **Efficiency:** MobileNet_V2 is optimized for mobile devices; its lower confidence is expected as it uses fewer parameters to reach a conclusion.